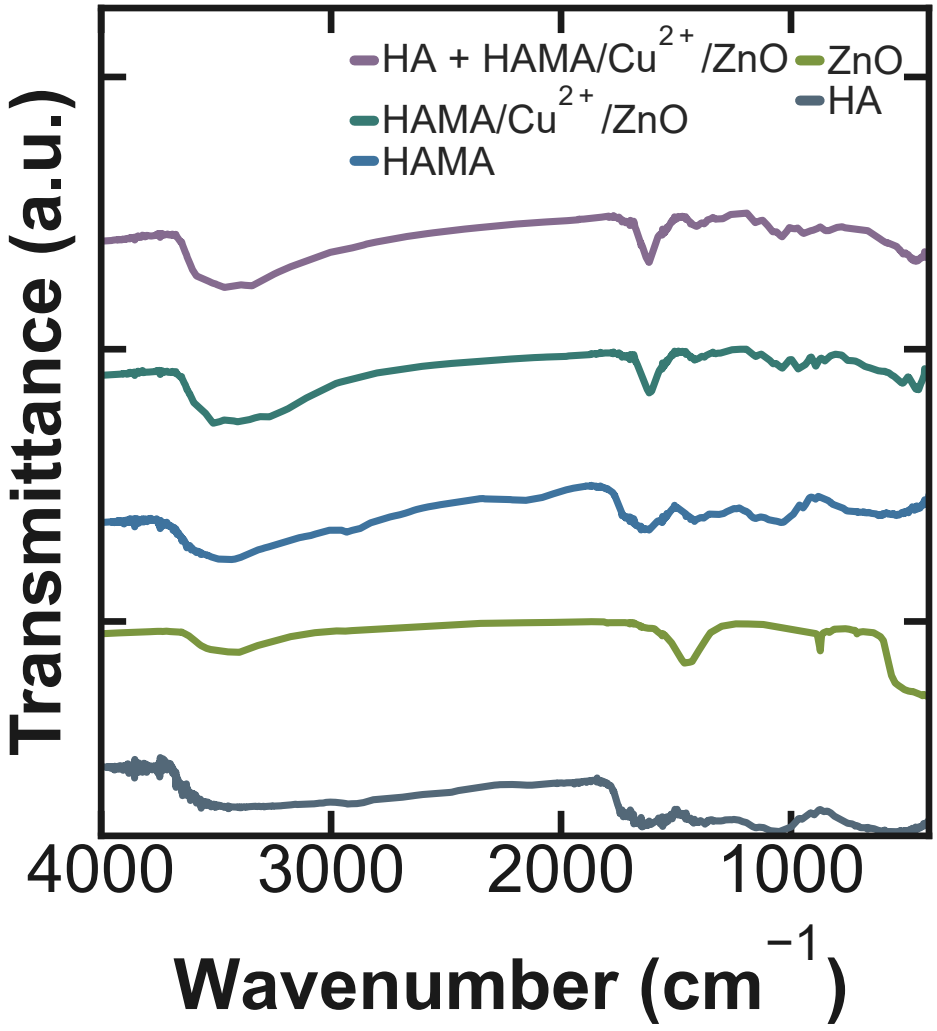
HA + HAMA/Cu2+ /ZnO
ZnO
HA
HAMA/Cu2+ /ZnO
HAMA
Transmittance (a.u.)
4000
3000
2000
1000
Wavenumber (cm−1)